

VerdictEdge // On-Device Legal AI

Air-Gapped Mobile Contract Risk Analyzer, Indian Statutory Engine & Multilingual Neural Voice

Status: Active /
Production-Ready

Platform: Native Android (Kotlin + Jetpack Compose)
Engine: Dual Architecture (Sherpa-ONNX + Google ML Kit)

Date: September
2026

0 KB

CLOUD DATA LEAKS

< 1.0s

ON-DEVICE LATENCY

3

LANGUAGES (EN/HI/KN)

100%

OFFLINE NEURAL VOICE

⚡ 1. Executive Summary & Core Mission

VerdictEdge is a privacy-first, 100% air-gapped mobile application engineered to analyze complex contracts, non-disclosure agreements (NDAs), and employment covenants directly on-device. By executing all optical character recognition (OCR), statutory evaluation under the *Indian Contract Act, 1872*, and neural text-to-speech narration on local smartphone hardware, VerdictEdge eliminates third-party cloud data leaks, requires zero subscriptions, and provides instant, plain-language legal audits.

🔄 2. End-to-End System Pipeline



🛡️ 3. Core Problems & VerdictEdge Solutions

⚠️ The Critical Problems

- **Confidentiality Leaks:** Uploading sensitive NDAs to commercial cloud LLMs (ChatGPT, Claude) risks trade secret breaches and client confidentiality violations.
- **Predatory Traps:** Unilateral non-competes, hidden recurring penalties, and unbounded indemnities often go unnoticed by non-lawyers.
- **The Vernacular Barrier:** In multilingual societies like India, contracts are drafted in archaic English legal jargon, disenfranchising vernacular speakers.

💡 The VerdictEdge Solutions

- **100% Air-Gapped Security:** Zero internet permissions required; all inference runs strictly in volatile RAM with zero data persistence.
- **Indian Statutory Voidability:** Instant alerts on terms rendered void under Indian jurisprudence (Sec 27 non-compete, Sec 28 litigation restraints).
- **Multilingual Neural Audio:** Studio-quality voice summaries in English, Hindi, and Kannada for on-the-go contract audits.

4. Indian Statutory & Contract Analysis Engine

The core engine evaluates agreements deterministically and semantically against codified Indian jurisprudence and commercial contract doctrines:

Module	Analytical Capability	Statutory Basis / Legal Doctrine
Statutory Voidability	Detects post-employment non-compete covenants and litigation jurisdiction restrictions.	Indian Contract Act, 1872: - Section 27 (Restraint of Trade / Void covenants) - Section 28 (Restraint of Legal Proceedings)
Financial Exposure	Flags uncapped late fees, recurring penalties, and indemnity triggers with exact monetary values.	Section 74: Liquidated damages vs unconscionable penalties doctrine.
Ambiguity Detector	Highlights subjective phrases ("at sole discretion", "reasonable efforts", "from time to time").	Identifies unilateral discretion and contract vagueness risks.
Problem-Solution Breakdown	Extracts the offending clause, explains the hazard, and auto-generates counter-offer clauses.	Ready-to-send contractual counter-proposals to equalize bargaining leverage.
Pre-Signing Checklist	Interactive verification list of resolved vs outstanding dealbreaker items.	Executive operational checklist ensuring pre-signing due diligence.

5. 100% Offline Multilingual Neural Voice (Sherpa-ONNX)

Sherpa-ONNX + Piper VITS Engine

Integrated `sherpa-onnx-static-link-onnxruntime-1.13.8.aar` paired with an INT8-quantized Piper VITS neural voice model (`vits-piper-en\_US-amy-medium-int8`) running locally via ONNX Runtime:

```
// Calibrated Acoustic Parameters
noiseScale = 0.72f // Natural emotion without sibilance
noiseScaleW = 0.85f // Balanced syllable durations
lengthScale = 1.02f // Crisp ~150 WPM legal narration pace
```

- **Digital Click Elimination:** 5ms soft linear PCM ramp-in/out removes audio initialization pop/clicks.
- **Legal Phonetic Normalization:** Normalizes citations (Sec. 27 → Section 27), legal Latin (vs. → versus, e.g. → for example), and reorders currency (INR 50000 → 50,000 Rupees).

Vernacular Routing & Prosody

Dual-Engine design routes languages dynamically to the optimal offline voice pipeline:

- **English:** Routed to Sherpa-ONNX Neural Engine with one-tap toggle to standard Android voice.
- **Hindi:** Routes to Google's neural `hi-in-x-hie-local` voice with relaxed cadence (0.88f) and Devanagari currency translation (₹ → रुपये, % → प्रतिशत, slashes → pauses).
- **Kannada:** Routes to `kn-in-x-knd-local` with regional legal articulation (₹ → ರೂಪಾಯಿ, % → ಪ್ರತಿಶತ).
- **Tab State Synchronization:** Switching language tabs instantly cancels active audio playback and resets control buttons cleanly.

6. Multilingual Localization Architecture

Full tri-lingual coverage ensures accessibility for English, Hindi (हिंदी), and Kannada (ಕನ್ನಡ) speakers across all risk summaries, statutory grounds, problem-solution breakdowns, checklists, and spoken audio summaries.

Visual verification captured directly from live test runs on Android 15 (Pixel 8 Pro emulator):

The image displays four distinct user interfaces for AI-powered legal document analysis:

- 1. Home Scanner & OCR:** The VerdictEdge interface features a header with a 'VerdictEdge' logo and a 'Gemini 2B Engine Active' status. Below the logo is a text input field for 'Enter text or scan a document to review'. A large text area for pasting contract clauses is present, along with 'Camera' and 'Gallery' buttons. At the bottom is a 'Generate Contract' button.
- 2. Critical Risk Analysis:** The 'Analysis Report' interface for 'Local offline Gemini 2B risk evaluation'. It highlights a 'CRITICAL RISK' in red, stating 'Critical liability exposures and unilateral terms detected'. Under 'Key Points to Address', it lists 'Uncapped Liability' (indemnification section lacks a financial liability ceiling), 'Broad Indemnity' (broad obligation covers indirect damages and third-party claims), and 'Unilateral Obligation' (obligation imposed exclusively on the receiving party). A 'Summary' section at the bottom mentions a 'FREELANCE DEVELOPMENT AGREEMENT'.
- 3. Sherpa Neural Audio:** The 'Analysis Report' interface for 'Export PDF'. It shows 'MODERATE RISK' with a warning about 'Unilateral and broad payment obligations, deadlines, and ambiguous discretionary terms'. It includes a 'Plain-Language Audio Summary' section and 'Key Deadlines & Obligations' for a '3 years' period, noting required notice periods and potential class action status.
- 4. Hindi Audio Summary:** The 'विश्लेषण रिपोर्ट' (Analysis Report) interface for 'Audio File'. It shows 'मध्यम जोखिम' (Moderate Risk) and 'मुख्य तिथियाँ और दायित्व' (Key Deadlines & Obligations) for a '3 years' period, mentioning a 'Standard Clause' and potential class action status.

Subsystem	Technology & Architecture	Offline & Security Profile
User Interface	Jetpack Compose (Material 3), Pitch-Black OLED UI (#0A0A0A)	Zero analytics tracking; zero cloud telemetry
Document Vision	Google ML Kit Text Recognition (Camera & Multi-PDF)	100% on-device vision inference
Contract Reasoning	Deterministic Statutory Engine + Regex Normalizer	Sub-second execution; zero cloud LLM dependencies
Neural Voice (EN)	Sherpa-ONNX JNI + ONNX Runtime (C++) + Piper VITS INT8	Embedded INT8 weights; 0 KB network calls
Indic Voice (HI/KN)	Android System TTS (Neural Voices: `hie`, `knd`)	On-device cached TTS engines
Report Export	Android PdfDocument Vector Generator	Local encrypted storage; instant print/share

- **Branch Status:** Packaged and committed on dedicated branch `UpdateVoices`.
- **Test Coverage:** 100% passing unit & instrumented tests (ContractEngineStressTest & OcrManagerTest).
- **APK Footprint:** Zero external runtime API dependencies; strict air-gapped security profile.

Sai Skanda (Team Lead): Initial architecture, core Android platform & on-device Gemma 2B LLM.

Syed Afnan: Voice transcription pipeline & Google ML Kit on-device OCR integration.

Shreesha K S: OCR pipeline bug fixes, stability & Sherpa-ONNX local neural TTS engine.

Samarth Sajjan: UI fixes, Jetpack Compose design system & frontend usability polish.